# Manuscript outline

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## Introduction

1. Phenological shifts with climate change
   (a) Shifts in spring/autumn modify start and end of season
   (b) Early spring + delayed autumn should increase growth
2. Positive relationship between season length and growth is important
   (a) Carbon cycle models rely on this positive relationship
   (b) This could impact climate solutions
3. Maybe longer GS don't increase growth
   (a) Dow showed longer GS didn't increase growth locally
   (b) Cabon showed decoupling between GPP/Growth
4. Longer GS does not necessarily mean more favorable conditions for growth throughout the season
   (a) Longer seasons are not necessarily warmer
   (b) Earlier SOS, also means earlier EOS—gains cool early season days (little increase in GDD) but miss warmer late season days (Figure 1)
   (c) Water availability may constrain growth
   (d) Competition may offset extra days to grow especially for below canopy trees
   (e) Length and speed of growth is physiologically (or genetically?) controlled
   (f) And how much phenology can track these changing conditions matters
5. Leaf and wood phenology are not symmetrically plastic
   (a) Leaf phenology in the spring is very plastic (well studied)
   (b) Leaf phenology and growth are synchronized = good proxy for growth SOS and EOS
   (c) However, this does not say anything how much trees grow in a year
   (d) How flexible growth is may change between species, but also across populations
6. Local adaptation to inform growth with climate change
   (a) Trees from northern provenance may grow less with longer seasons
   (b) Budset is strongly locally adapted
   (c) Common gardens are important for climate change and tree growth
7. Tree growth will vary across species but will also depend on future environmental conditions
   (a) To understand GS and growth relationship, need to look more precisely at when trees grow during the GS
   (b) If growth is not plastic enough, trees may not be able to track favorable changing conditions in the season

(c) Varies across species and population, with the latter having a potential to inform how species will respond to future conditions

(d) Juvenile trees store little carbon, but are crucial for forest regeneration capacity

8. Answer how GS and growth relate using juvenile trees in a common garden

    (a) Disentangle GS length vs GS thermal conditions

    (b) Growth with tree rings, SOS and EOS with leaf phenology

    (c) Calendar season: days between SOS and EOS and thermal season: GDD between these days

    (d) Will look at different species from different provenances

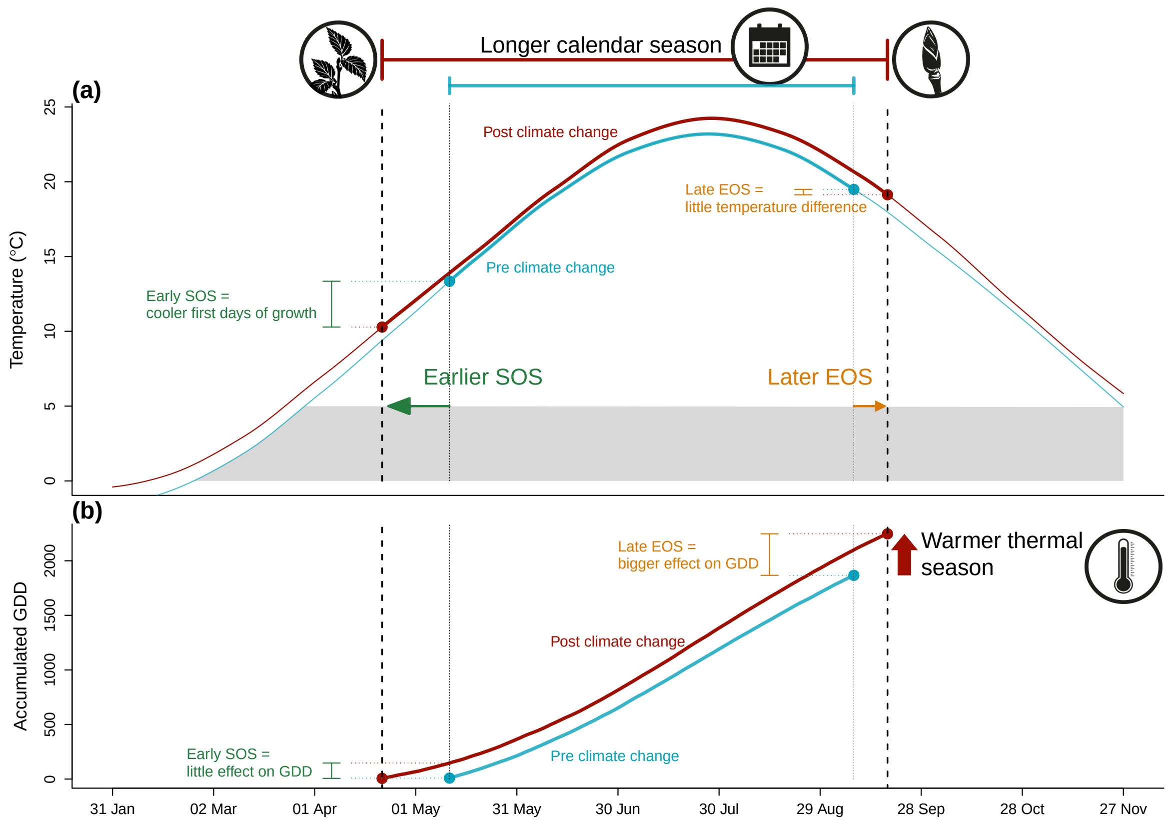


Figure 1: We show how shifting growing season length with climate change affects both the temperature trees experience throughout the growing season (a), and how much growing degree days (GDD) they accumulate (b). We used mean daily temperature data from the Boston airport Climate station for the smoothed temperature and GDD curves. We represent the changes in temperature before (1945-1965 in blue) and after the 1980s increase in climate warming (2005-2025 in red). With climate change, spring events (green arrow) have advanced and fall phenology (orange arrow) has delayed, but to much lesser extent (day advancements are fake and represented by the arrows), this lengthened the calendar growing season, but concurently changed the temperature trees experience early in their season. Growth onset earlier in the season means the first days of growth are cooler (a) resulting in little accumulation of GDD (b). Though shifts in EOS are weaker and the temperature experienced changed little (a), the GDD accumulation rate late in the season is higher than early in the season. Alongside warmer summer temperature, this results in higher GDD accumulation, warming their thermal seasons.

# Results

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
   (b) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
   (c) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
   (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
   (e) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
   (f) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S5).
2. Does tree growth increase with longer growing seasons?
   (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
   (b) Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.15$, $UI_{90}$ [0.06, 0.23]), *B. populifolia* ($\beta = 0.07$, $UI_{90}$ [0.02, 0.12]), and *B. alleghaniensis* ($\beta = 0.04$, $UI_{90}$ [0, 0.08]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.02$, $UI_{90}$ [-0.01, 0.05]) (Figure 2a and e) (TableS1).
   (c) *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, $UI_{90}$ [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, $UI_{90}$ [0.05, 0.31]).
   (d) *B. alleghaniensis* ($\beta = 0.03$, $UI_{90}$ [-0.03, 0.09]), and *B. populifolia* ($\beta = 0.05$, $UI_{90}$ [-0.02, 0.13]) trended in the same direction (Figure 2b and f) (Table S1).
   (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S4)
   (f) On a scaled axis, both predictors were similarly predictive (FigureS6).
3. Did a later start and end of season increase growth?
   (a) A longer calendar season increased growth for two species, but only one grew more with an earlier start of season (*A. incana*: $\beta =$ -0.07, $UI_{90}$ [-0.13, -0.01]) (Figure 2c and g) (Table S1).
   (b) *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction ($\beta = 0.1$, $UI_{90}$ [0, 0.2]) (Figure 2c and g) (Table S1). git statusgit status
   (c) Three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, $UI_{90}$ [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, $UI_{90}$ [0.02, 0.17]), with *B. papyrifera* ($\beta = 0.21$, $UI_{90}$ [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 2d and h) (Table S1).
   (d) The estimates were unchanged when using the full (SOS $n$ = 239 and $n$ = 230) or restricted datasets ($n$ =227) (Figure S1a).

4. How did tree growth change across years, provenances and tree ids?

    (a) We did not find any clear effect of year on growth (Figure **??**) and there is no clear differences of growth looking across species (Figure S2 and Figure S3)

    (b) Growth did not change across any provenances (Figure 3) (Table S2), and the variation across sites ($\sigma$ = 0.17, $UI_{90}$ [0.02, 0.48) and tree ids ($\sigma$ = 0.17, $UI_{90}$ [0.05, 0.27]) was similar

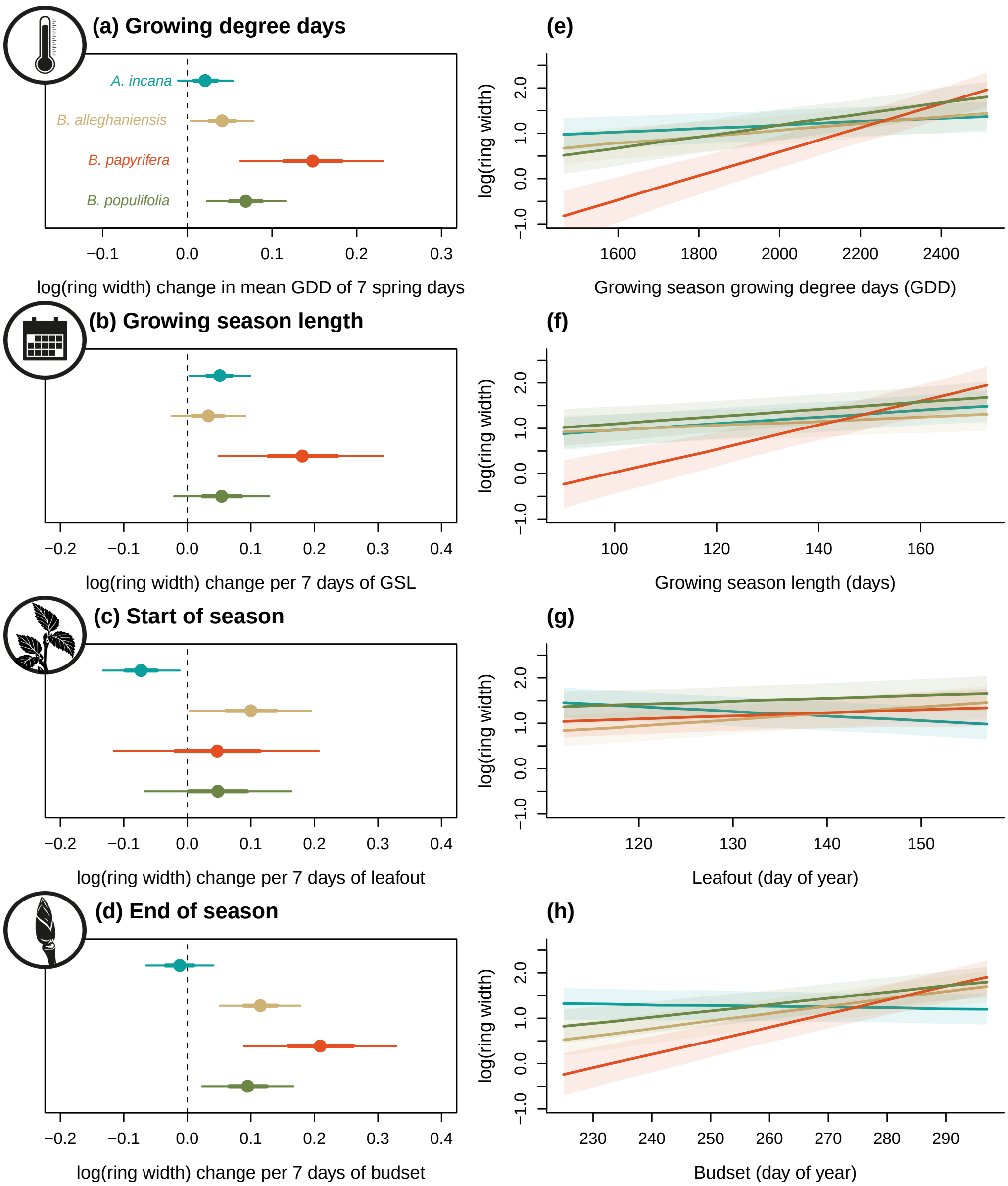


Figure 2: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (55.73) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

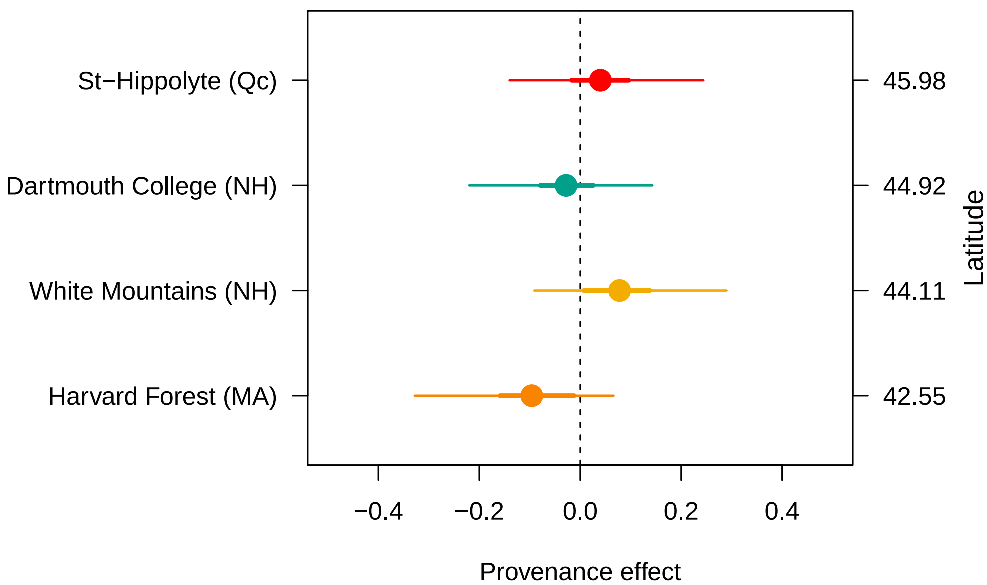


Figure 3: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

# Discussion

1. (a) Used 81 juvenile trees of 4 species from 4 provenances to understand if longer GS increased growth
      i. We used annual growth with tree rings and leaf phenology for SOS and EOS
      ii. Longer GS increased growth for all species, but varied depending on the metric
      iii. Thermal season was a better predictor for growth than the calendar season
      iv. Little effect of provenance

2. The thermal season better predicts growth than the calendar season
   (a) Thermal is a better predictor of growth because it's more closely aligned to when and how fast trees grow
      i. Thermal weights growth days better than calendar season
      ii. Cool spring days count for less for thermal season vs calendar season (Figure 1)
      iii. Stronger response to EOS than SOS, because temperature is warmer = more GDD (Figure 1) (Buonaiuto, unpublished)
   (b) SOS and EOS to disentangle the calendar season effect on growth
      i. Early SOS didn't increase growth for most species (like Dow 2022)
      ii. Studies that solely focus on SOS will miss that longer seasons may lead to more growth
      iii. Late EOS increased growth for most species; relates to GDD, less to GSL
   (c) Future response of growth in response to soil moisture is uncertain
      i. Later EOS coupled with appropriate soil moisture = growth increase—as we and others found
      ii. No current increased drought in our species native range = growth would increase
      iii. Positive effect of later EOS offset if soil moisture becomes inadequate
      iv. While 2020 had a late of season drought, late of season droughts in the north east are rate—could change with warmer conditions
      v. Other temperate regions have growth decrease attributed to reduced soil moisture

3. Contrasting growth responses across species, but not across provenances
   (a) Species responded differently to different season predictors
      i. *A. incana* grow more with GSL and earlier SOS: most growth happens in spring
      ii. *A. incana* is ruderal, takes advantage of cool and short days in spring
      iii. Contrary, *B. papyrifera* and *B. populifolia* driven by EOS: more reactive to warmer, late season temperature
      iv. *B. alleghaniensis* SOS and EOS trends counterinteract = no response to GSL
   (b) Growth strategies across species may change how responsive they are
      i. All our species are either fast or moderately fast growing species—known to track changing conditions better
      ii. This could change if we used slow growing species
      iii. Juvenile stages are known to be more reactive, could explain why they grew more with warmer seasons
      iv. These trees in nature would be below canopy, so they would have much less light and might not have responded positively
   (c) No effect of provenance suggests local adaption effect on growth is weak under current conditions
      i. Good understanding of which leaf phenophase is locally adapted: budset > leafout
      ii. Little evidence of local adaptation on growth may be caused by little local adaptation on budset
      iii. We know little about the effect of local adaptation on growth
      iv. Understanding this requires teasing out the effects of local adaptation on leaf and wood phenology at the SOS and EOS

4. Future carbon models and range shift projections need to consider species

(a) What we did is important

  i. Juvenile trees may be more responsive, but they would be competing for light in nature
  ii. EOS and SOS asymmetrically matter to growth and we don't understand why
  iii. Highlights importance of considering not just SOS (e.g. Dow, Wheeler2016Snow)
  iv. Closely related species trends differed; they matter and we don't consider them enough

# Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

| Species | GDD | GSL | SOS | EOS |
|---|---|---|---|---|
| *A. incana* | 0.02 [-0.01, 0.05] | 0.05 [0.00, 0.10] | -0.07 [-0.13, -0.01] | -0.01 [-0.07, 0.04] |
| *B. alleghaniensis* | 0.04 [0.00, 0.08] | 0.03 [-0.03, 0.09] | 0.10 [0.00, 0.20] | 0.12 [0.05, 0.18] |
| *B. papyrifera* | 0.15 [0.06, 0.23] | 0.18 [0.05, 0.31] | 0.05 [-0.12, 0.21] | 0.21 [0.09, 0.33] |
| *B. populifolia* | 0.07 [0.02, 0.12] | 0.05 [-0.02, 0.13] | 0.05 [-0.07, 0.16] | 0.10 [0.02, 0.17] |

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

| Site | GDD | GSL | SOS | EOS | Lon. |
|---|---|---|---|---|---|
| Harvard Forest (MA) | -0.10 [-0.33, 0.07] | -0.09 [-0.34, 0.09] | -0.04 [-0.26, 0.12] | -0.10 [-0.33, 0.07] | -72.19 |
| White Mountains (NH) | 0.08 [-0.09, 0.29] | 0.08 [-0.09, 0.31] | 0.08 [-0.08, 0.29] | 0.07 [-0.09, 0.27] | -71.23 |
| Dartmouth College (NH) | -0.03 [-0.22, 0.14] | -0.04 [-0.25, 0.15] | -0.06 [-0.27, 0.10] | -0.04 [-0.24, 0.12] | -71.15 |
| St-Hippolyte (Qc) | 0.04 [-0.14, 0.24] | 0.04 [-0.16, 0.26] | 0.02 [-0.15, 0.22] | 0.04 [-0.14, 0.24] | -74.03 |

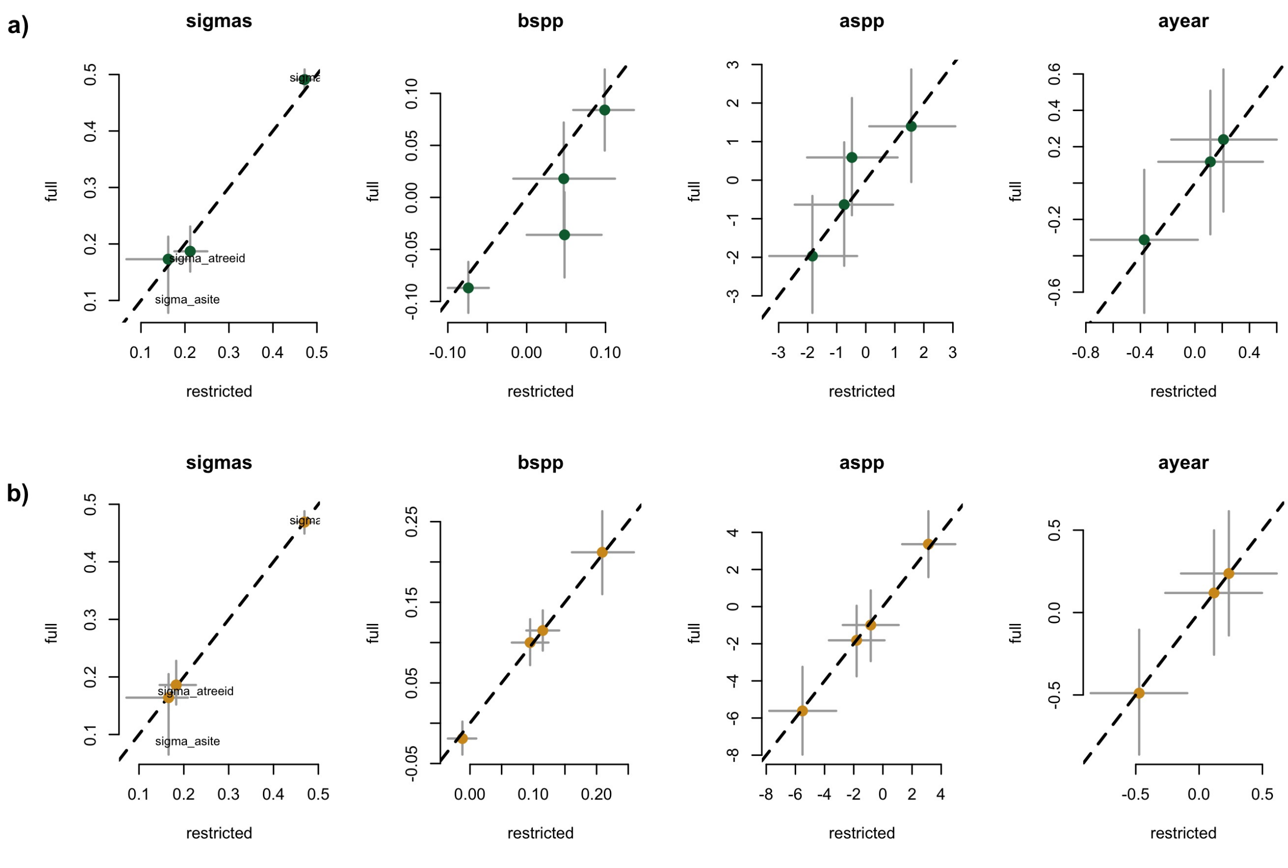


Figure S1: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

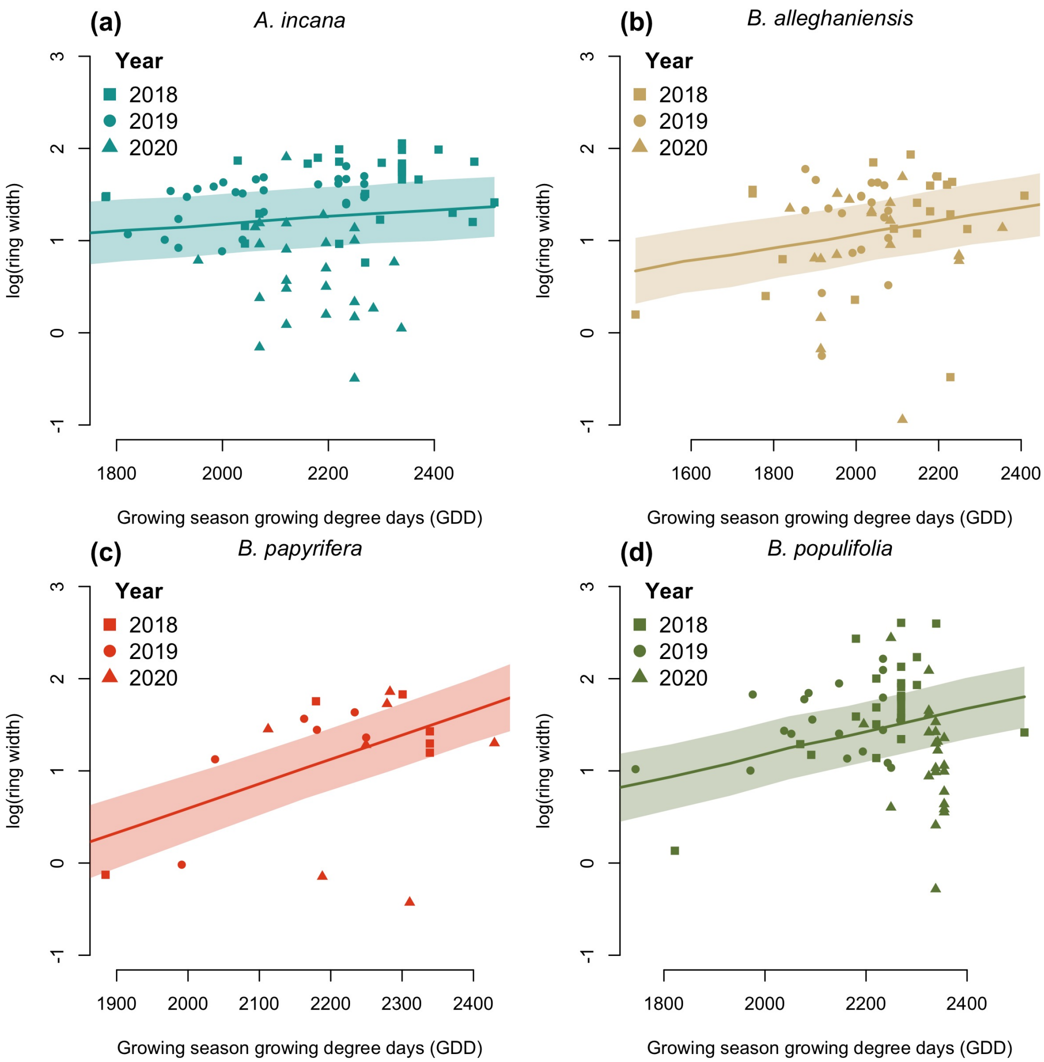


Figure S2: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

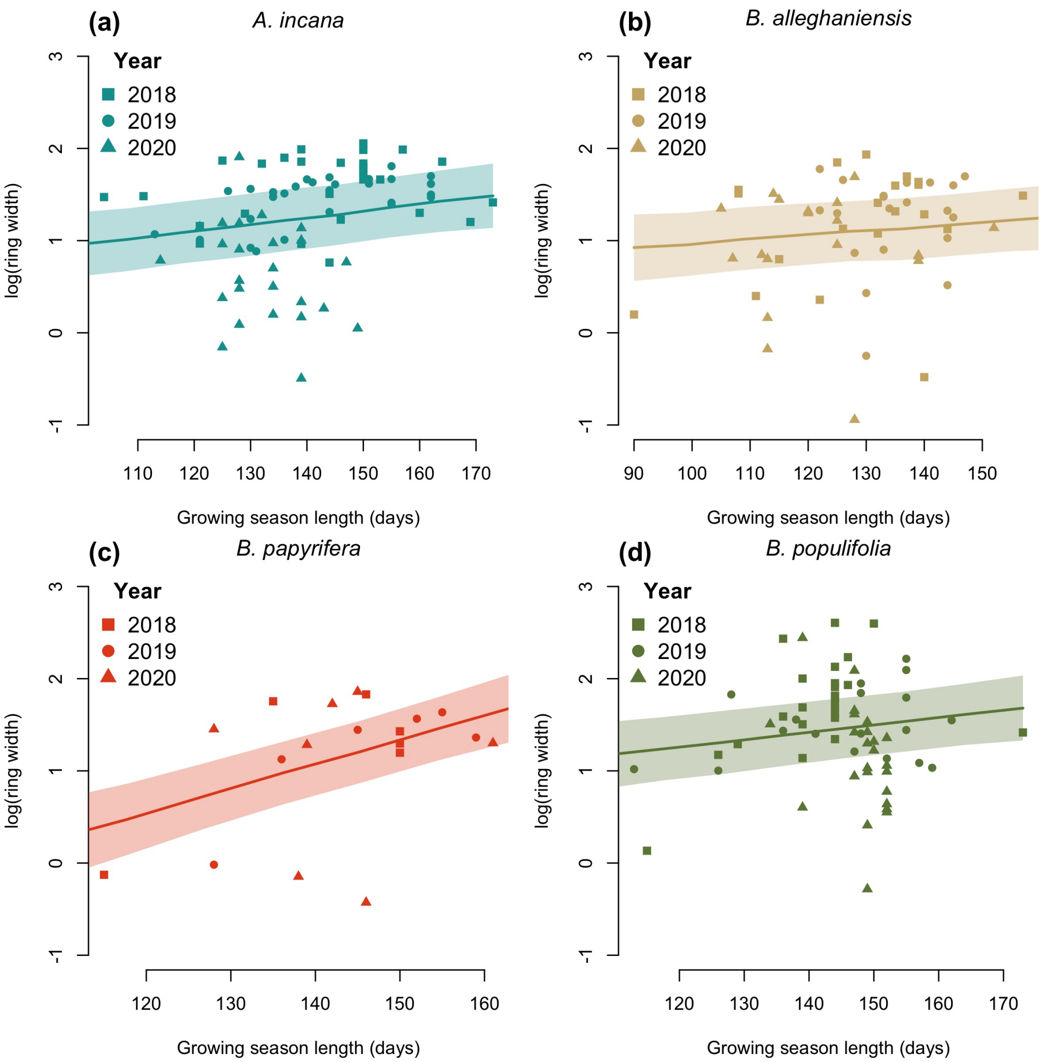


Figure S3: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

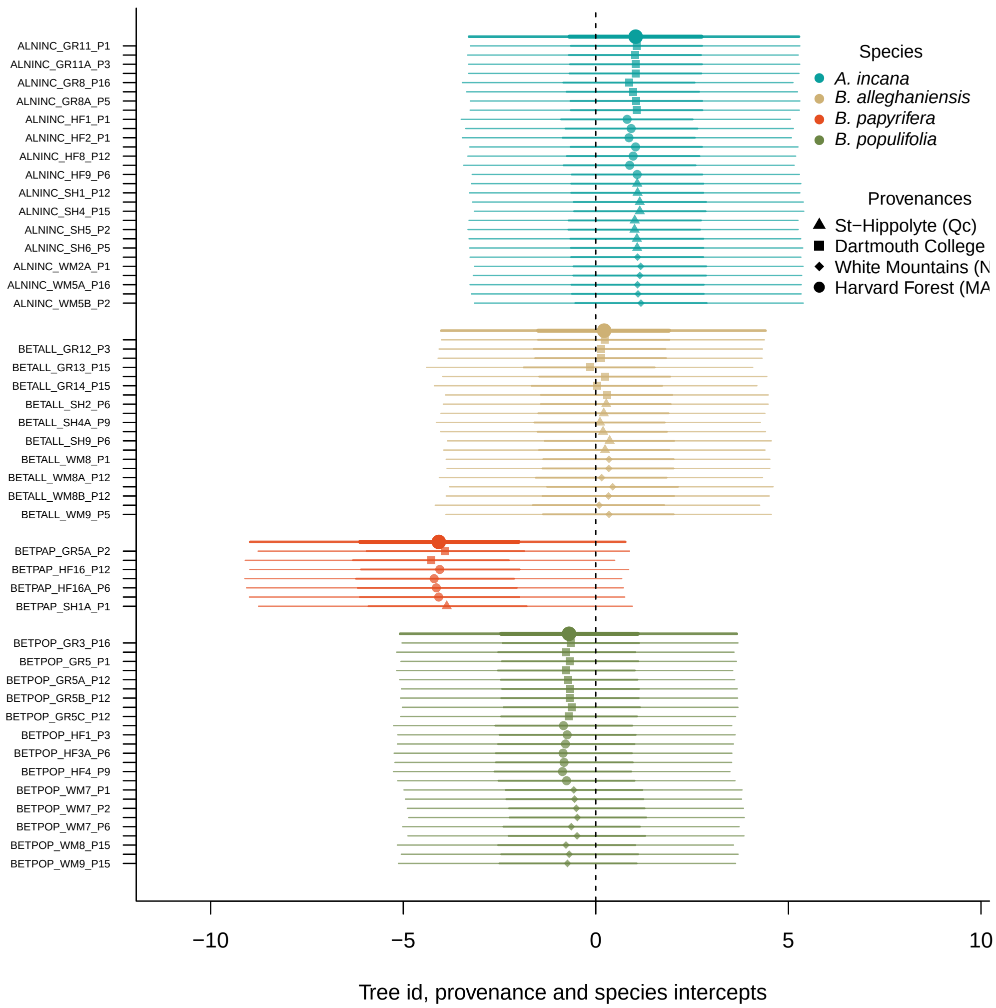


Figure S4: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

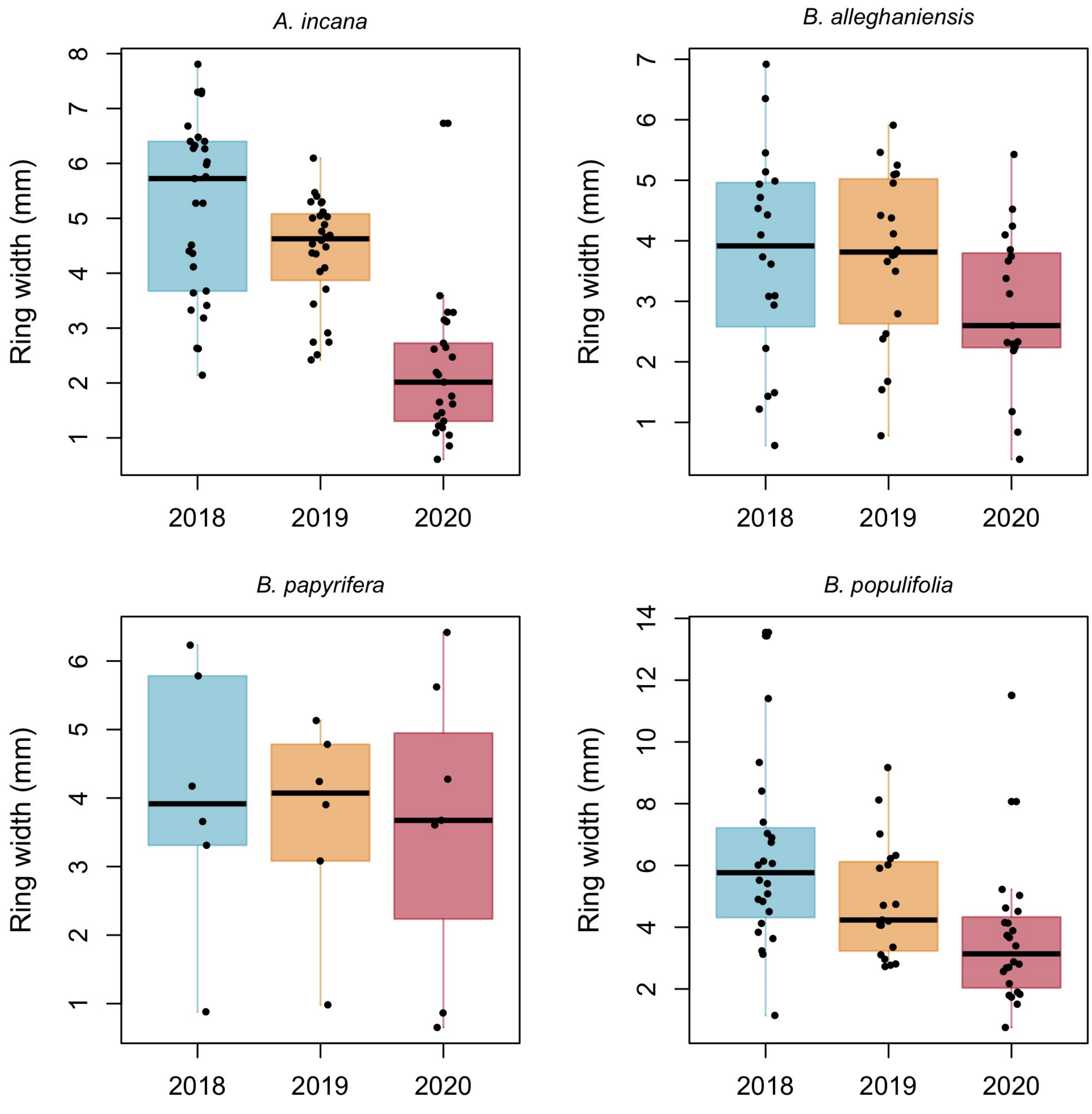


Figure S5: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

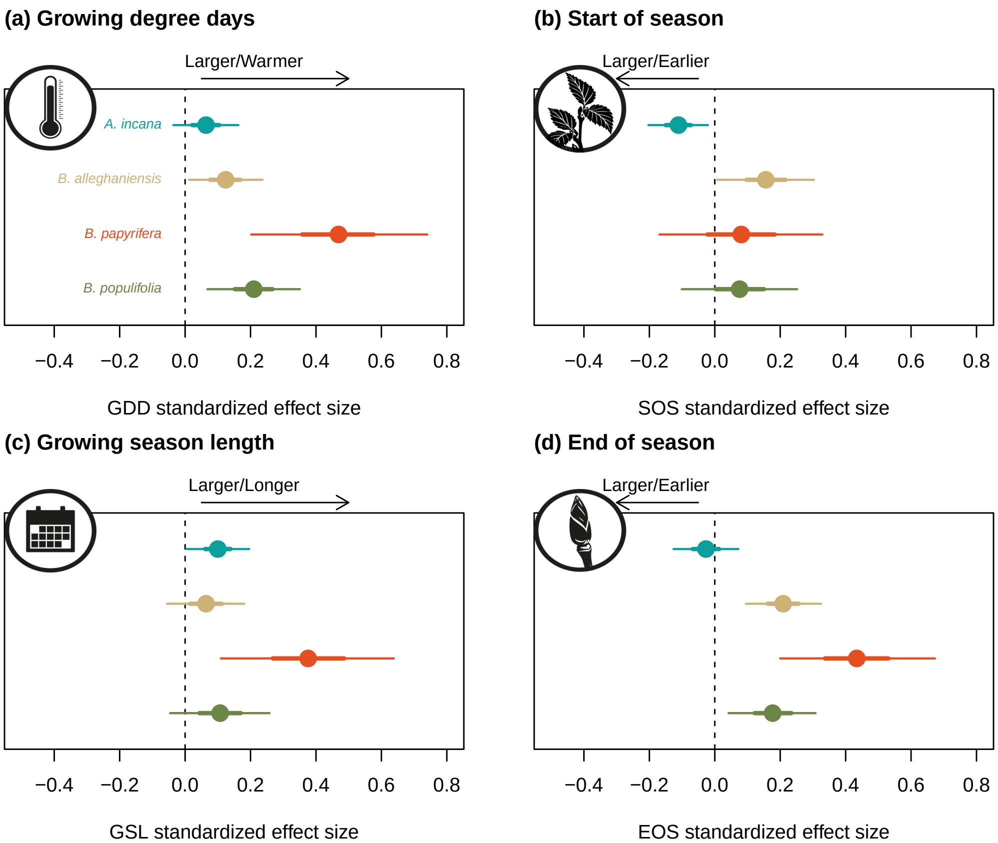


Figure S6: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters across predictors

| Parameter | GDD | GSL | SOS | EOS |
|---|---|---|---|---|
| $\sigma_{\alpha_{tree}}$ | 0.17 [0.05, 0.27] | 0.18 [0.06, 0.28] | 0.21 [0.10, 0.30] | 0.18 [0.06, 0.28] |
| $\sigma_{\alpha_{site}}$ | 0.17 [0.02, 0.47] | 0.18 [0.02, 0.48] | 0.17 [0.02, 0.46] | 0.17 [0.02, 0.46] |
| $\sigma_y$ | 0.48 [0.43, 0.53] | 0.48 [0.44, 0.53] | 0.47 [0.43, 0.52] | 0.47 [0.42, 0.52] |
| $\beta_{spp_{A.incana}}$ | 0.02 [-0.01, 0.05] | 0.05 [0.00, 0.10] | -0.07 [-0.13, -0.01] | -0.01 [-0.07, 0.04] |
| $\beta_{spp_{B.alleghaniensis}}$ | 0.04 [0.00, 0.08] | 0.03 [-0.03, 0.09] | 0.10 [0.00, 0.20] | 0.12 [0.05, 0.18] |
| $\beta_{spp_{B.papyrifera}}$ | 0.15 [0.06, 0.23] | 0.18 [0.05, 0.31] | 0.05 [-0.12, 0.21] | 0.21 [0.09, 0.33] |
| $\beta_{spp_{B.populifolia}}$ | 0.07 [0.02, 0.12] | 0.05 [-0.02, 0.13] | 0.05 [-0.07, 0.16] | 0.10 [0.02, 0.17] |
| $\alpha_{site_{DartmouthCollege(NH)}}$ | -0.03 [-0.22, 0.14] | -0.04 [-0.25, 0.15] | -0.06 [-0.27, 0.10] | -0.04 [-0.24, 0.12] |
| $\alpha_{site_{HarvardForest(MA)}}$ | -0.10 [-0.33, 0.07] | -0.09 [-0.34, 0.09] | -0.04 [-0.26, 0.12] | -0.10 [-0.33, 0.07] |
| $\alpha_{site_{St-Hippolyte(Qc)}}$ | 0.04 [-0.14, 0.24] | 0.04 [-0.16, 0.26] | 0.02 [-0.15, 0.22] | 0.04 [-0.14, 0.24] |
| $\alpha_{site_{WhiteMountains(NH)}}$ | 0.08 [-0.09, 0.29] | 0.08 [-0.09, 0.31] | 0.08 [-0.08, 0.29] | 0.07 [-0.09, 0.27] |
| $\alpha_{year_{2018}}$ | 0.19 [-0.77, 1.15] | 0.24 [-0.70, 1.17] | 0.20 [-0.75, 1.14] | 0.22 [-0.71, 1.16] |
| $\alpha_{year_{2019}}$ | 0.18 [-0.79, 1.14] | 0.10 [-0.83, 1.06] | 0.11 [-0.84, 1.04] | 0.11 [-0.83, 1.05] |
| $\alpha_{year_{2020}}$ | -0.43 [-1.40, 0.53] | -0.34 [-1.27, 0.60] | -0.38 [-1.32, 0.56] | -0.49 [-1.42, 0.46] |
| $\alpha_{spp_{A.incana}}$ | 1.06 [-3.19, 5.21] | 0.11 [-3.92, 4.21] | 1.48 [-2.13, 5.13] | 3.06 [-1.40, 7.36] |
| $\alpha_{spp_{B.alleghaniensis}}$ | 0.21 [-3.97, 4.34] | 0.35 [-3.69, 4.53] | -1.91 [-5.64, 1.79] | -1.85 [-6.39, 2.59] |
| $\alpha_{spp_{B.papyrifera}}$ | -4.00 [-8.89, 0.75] | -2.61 [-7.17, 1.95] | -0.79 [-5.03, 3.36] | -5.57 [-11.02, -0.07] |
| $\alpha_{spp_{B.populifolia}}$ | -0.66 [-4.94, 3.63] | 0.22 [-3.97, 4.45] | -0.53 [-4.29, 3.22] | -0.88 [-5.38, 3.67] |

Table S4: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

| Parameter | Estimate |
| --- | --- |
| $\sigma_{\alpha_{tree}}$ | 0.52 [0.41, 0.64] |
| $\sigma_{\alpha_{site}}$ | 0.27 [0.03, 0.73] |
| $\sigma_y$ | 0.63 [0.57, 0.70] |
| $\beta_{spp_{A.incana}}$ | 0.11 [-0.03, 0.25] |
| $\beta_{spp_{B.alleghaniensis}}$ | 0.20 [0.02, 0.36] |
| $\beta_{spp_{B.papyrifera}}$ | 0.92 [0.49, 1.34] |
| $\beta_{spp_{B.populifolia}}$ | 0.41 [0.20, 0.61] |
| $\alpha_{site_{DartmouthCollege(NH)}}$ | -0.11 [-0.47, 0.17] |
| $\alpha_{site_{HarvardForest(MA)}}$ | -0.09 [-0.45, 0.19] |
| $\alpha_{site_{St-Hippolyte(Qc)}}$ | 0.06 [-0.23, 0.42] |
| $\alpha_{site_{WhiteMountains(NH)}}$ | 0.15 [-0.11, 0.52] |
| $\alpha_{year_{2018}}$ | -0.36 [-1.32, 0.61] |
| $\alpha_{year_{2019}}$ | 0.33 [-0.65, 1.32] |
| $\alpha_{year_{2020}}$ | -0.04 [-1.01, 0.94] |
| $\alpha_{spp_{A.incana}}$ | 3.23 [-2.25, 8.55] |
| $\alpha_{spp_{B.alleghaniensis}}$ | 2.09 [-3.40, 7.54] |
| $\alpha_{spp_{B.papyrifera}}$ | -7.59 [-14.84, -0.29] |
| $\alpha_{spp_{B.populifolia}}$ | -0.07 [-5.98, 5.57] |

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